

Code Optimisation

The endless loop of rewriting code

Mithun Mohandas |
 mithun@digit.in

As a developer, we all want our code to be optimised the very first time we write a module. As an experienced developer, we want our code to just work the very first time we write a module. Jokes aside, code optimisation is all about writing software in such a manner that you minimised the compute and network usage. A bloated code does no good and each line of code is like a drop in the ocean. Put together, they end up being a massive load on compute resources. The end product becomes sluggish to use and you'll end up with irritated customers or worse, with none at all.

Take for example, the Windows 10 Start Menu. When your OS is installed on an SSD, that thing is very responsive even when you have a lot of software installed on your computer. However, if you were to be running Windows on a hard drive, then the Start Menu ends up more and more unresponsive as you install more software. It's astonishing that the Start Menu, which is such a focal part of the operating system is in such a state. This is also because developers, like the rest of us, tend to take for granted the sheer amount of performance and memory that's available. These were major constraints up until a few years ago and a lot of the software had to be optimised in order to ensure a passable experience. The

way things are now, we've skipped optimisation altogether.

Sometimes, optimisation also means to write less code or to write elegant code, as the purists put it. In practise, a lot of the time, the elegant code and the non-elegant code end up the same once they pass through the compiler. In such a situation, you've done nothing but massage your own ego. You've written an elegant but cryptic code that's understood by very few people in the world, and the next person who has to read and comprehend your code is going to curse the heck out of you and your forefathers. That said, a code that has fewer lines has fewer chances for error.

AN EXAMPLE

Here's an example of writing the same thing in two different ways, one is a little explicit while the other is elegant.

Non-elegant but explicit:

```
<div class=Section1>
    <table border="0" cellpadding="0" cellspacing="2" width="102" align="center">
        <tbody>
            <tr>
                <td width="80" align="center"><a href="sm3.htm"></a></td>
                <td width="80" align="center"><a href="booktoc.html"></a></td>
                <td width="80" align="center"><a href="sm5.htm"></a></td>
            </tr>
        </tbody>
    </table>
    <p class=chapter>Chapter 4: Deduction and Logic</p>
    <table border=0 cellspacing=3 cellpadding=0 align=right width=240>
        <tr>
            <td width=240 valign=top class="Normal">
                <p style=' ' > </p>
            </td>
        </tr>
    </table>
    <p class=quote>"The supreme task of the physicist is to arrive at those universal elementary laws from which the cosmos can be built up by pure deduction. There is no logical path to these laws; only intuition, resting on sympathetic understanding, can lead to them." [Einstein, 1879-1955]</p>
    <p class=quote>"From a drop of water," said [Sherlock Holmes], 'a logician could infer the possibility of an Atlantic or a Niagara without
```